Roll No. | | | | | | | |

SET-1

Odd sem End Semester Examination 2025

Name of the Course: B. Tech Biotechnology Semester: 5th

Name of the Paper: Fermentation Technology Paper Code: **TBT-503**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Each sub question carries 10 marks

Q1	(20marks)	
(a)	Discuss 5 types of the products obtained in Fermentation Industry.	
(b)	Describe methods for preservation of industrial microorganisms and their importance.	CO1
(c)	Outline the major milestones in the development of fermentation technology.	
Q2	(20 marks)	
(a)	Derive an expression for Energy Balance. Describe the phases of microbial growth in batch culture with a graph	
(b)	Derive an expression for total and instantaneous Biomass in Fed Batch culture assuming quasi steady state conditions.	CO2
(c)	Explain following terms- Inoculum development, Cryptic growth, secondary metabolites, Dilution rate, Monod Equation	
Q3	(20 marks)	
(a)	Compare thermal death kinetics and continuous sterilization processes.	CO3
(b)	How do you sterilize air? Compare Fibrous and Absolute air filters.	
(c)	How will you apply Plackett Burman for media designing if you are given 17 variables of media?	
Q4	(20 marks)	
(a)	Differentiate between Feedback Inhibition and Feedback Repression. Give any 3 types of Feedback Inhibition mechanisms.	CO4
(b)	How can you promote overproduction of Ornithine (Primary metabolite) and IMP nucleotides? Explain with process flowcharts.	
(c)	Discuss role of cyclic AMP and excess glucose concentration in Lac operon regulation? What will happen if glucose is not present and Lactose is there?	
Q5	(20 marks)	
(a)	Write short note on Microbial Leaching and different processes involved in it.Give 2 examples.	CO5
(b)	Explain fermentative production and estimation of Citric acid.	
(c)	How do you produce Tetracycline? Give its 2 applications.	